

Aravind Gudikandula

aravind.g1942@gmail.com | +1-669-499-8490 | LinkedIn | Github

SUMMARY

AI/ML Engineer with experience in designing and deploying intelligent systems spanning computer vision, LLM integration, and RAG architectures. Hands-on expertise in PyTorch, LangChain, Gemini, and cloud-native MLOps pipelines on AWS and GCP, with a strong foundation in full-stack development to deliver end-to-end production-ready AI applications. Passionate about pushing the boundaries of generative AI and multimodal systems to solve real-world problems at scale.

EXPERIENCE

- San Jose State university** Remote
Research Assistant Mar 2026 - Present
 - Built a high-performance hybrid pipeline combining **YOLOv8** with **Stable Diffusion 3** and **InstructPix2Pix**, enabling real-time object detection and seamless object add/remove/replace editing through an intuitive web interface.
 - Built an adaptive parameter-tuning system that automatically modifies diffusion steps (38–55), guidance scales (8.0–9.2), and mask expansions (12%–24%) according to object size, achieving photorealistic blending and uniform texture quality.
 - Engineered a low-latency system with **FastAPI + React (TypeScript)**, MD5-based smart caching, and advanced **OpenCV/PIL** post-processing for color transforms, weather effects, and multi-seed scoring to auto-select the best generative output.
- Hitachi Vantara** Remote
ML Engineer Jan 2023 - Dec 2023
 - Developed and deployed NLP pipelines using HuggingFace Transformers and PyTorch for document classification and entity extraction, integrating GPT-3.5 via OpenAI API to automate internal enterprise knowledge workflows.
 - Built an early-stage RAG system using LangChain and Pinecone vector database to power an internal enterprise Q&A assistant, reducing manual document lookup time by 35% across 1M+ indexed records.
 - Designed MLflow-based experiment tracking and model versioning pipelines for fine-tuned BERT models, improving reproducibility and reducing model deployment cycles from weeks to days via automated CI/CD integration.
 - Collaborated with cross-functional teams to productionize ML models via REST APIs (FastAPI), monitoring model drift and retraining triggers using custom evaluation metrics across multiple high-priority enterprise releases.
- F13 Technologies** Remote
AWS Intern Feb 2022 - Nov 2022
 - Designed and deployed cloud infrastructure on AWS (EC2, S3, Lambda, RDS, CloudFormation) supporting 5+ scalable applications with 99.8% availability, leveraging infrastructure-as-code practices.
 - Automated deployment workflows and CI/CD pipelines using CloudFormation templates and Python scripts, decreasing deployment time by 30% and minimizing production incidents through repeatable automation.
 - Analyzed cloud resource utilization with CloudWatch and Cost Explorer, identifying optimization opportunities that delivered 18% cost savings in monthly AWS expenditure while maintaining performance targets.
 - Migrated legacy applications to cloud-native architectures, improving system scalability and reducing infrastructure maintenance overhead by 25% through systematic problem-solving and technical documentation.

PROJECTS

- Dynamic Scene Generation Framework for Smart Vehicles** Jan 2025 - Dec 2025
Skills: Python, PyTorch, YOLOv8, Stable Diffusion, CLIP, FAISS, CUDA
 - Architected production-ready computer vision framework integrating Detection, Search, and Generative Insertion modules, exceeding targets with PSNR of 32.44 dB (29% above) and SSIM of 0.9226.
 - Streamlined ML pipeline implementing batch processing and FP16 mixed precision, achieving 12-24x speedup that accelerated processing from 6 hours to 15 minutes with 22 FPS at 95% accuracy.
- Medical Domain RAG System with LLM Integration** Dec 2024
Skills: BioMistral, Qdrant, LangChain, FastAPI, Llama.cpp, PubMedBERT, Python
 - Established open-source RAG system leveraging BioMistral (7B) and PubMedBERT, achieving 85% relevance accuracy on healthcare queries with local LLM deployment for data privacy.
 - Enhanced embedding pipeline using Qdrant vector database and LangChain with FastAPI integration, maintaining sub-second response times while decreasing overhead by 40%.

EDUCATION

- M.Sc Software Engineering** Jan 2024 - Dec 2025
San Jose State University GPA - 3.5
- B.Tech Information Technology** August 2019 - May 2023
Malla Reddy College of Engineering and Technology GPA - 3.7

SKILLS SUMMARY

- Frontend Development:** React, Angular, Vue.js, JavaScript, TypeScript, HTML5, CSS3, REST APIs, Responsive Design
- Backend Development:** Java, Python, C++, C, Golang, Ruby, Node.js, Express, SQL, REST APIs, Networking, Data Structures & Algorithms
- Gen AI & LLM Technologies:** GPT-4, Gemini API, OpenAI, Copilot, LLaMA, RAG, LangChain, Prompt Engineering, LLM Fine-tuning, LLM Integration, Data Analytics, Data Visualisation
- ML & MLOps:** PyTorch, TensorFlow, Scikit-learn, MLOps Pipelines, Model Deployment, YOLOv8, OpenCV, Model Fine-tuning, NLP, Neural Networks, Data Mining, Data Processing, Statistics, Data Science
- Cloud & Data Platforms:** AWS (EC2, S3, Lambda, RDS, CloudFormation, DynamoDB, ECS, SQS), GCP (Cloud Run/Build), Azure, MySQL, PostgreSQL, MongoDB, PowerBI
- DevOps Tools:** Docker, Kubernetes, Terraform, CI/CD, Jenkins, Git, Linux, JIRA, Confluence, Agile/SDLC
- Professional Skills:** Technical Skills, Authorization, Development, Communication Skills, Leadership, Innovation